

Key Component List

The parts whose numbers feed the software, the parameters and the simulator, with where each number came from

Document	BOATY-KCL-001
Issue	Issue C (for review)
Date	30 September 2026
Status	For review. One owner decision (CR-04) before the ADD, ICD and SSS pick it up.
Basis	ADD Issue E, ICD Issue D, SSS Issues A-C, FMEA Issue C
Content	17 key components, 11 findings, 19 datasheet entries

Document control

Issue	Date	Change	By
A	28 September 2026	First issue.	Claude (drafted)
B	29 September 2026	CR-04 accepted: SpeedyBee F405 WING APP is the baseline. Section 10 now records where each change went.	Claude, owner decision
C	30 September 2026	CR-08: the conditional antenna pole kit is outside the cap, in the field-kit budget.	Claude, owner decision

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: check the proposed parts against what you can actually buy, and the cost estimates against real prices. Every value is tagged with its source; orange-tagged values are guesses until measured.

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1. Purpose and how to read this list

The simulator and the software need exact numbers: which UART the mission computer is on, which ADC pin senses the motor rail, how fast an ESC stops when its signal disappears, how much thrust a pod gives. Until now the specifications said 'F405 class' and 'M10 GNSS'. This list turns those into specific parts and records the numbers that matter from each.

A **key component** is one whose characteristics feed the software, the ArduPilot parameters, the simulator's physics or an interface definition. Foam, glands and fasteners are not key components and stay in the ADD's BOM.

1.1 Where the numbers come from

Each value carries a source tag. The helm's numbers come mostly from ArduPilot's own board definition: the firmware we flash is compiled from those files, so they're more authoritative than a vendor PDF.

Tag	Meaning	Trust
AP	ArduPilot source. Read from ArduPilot's board definition, driver or parameter source at the commit below, and archived in docs/kcl/sources/. The firmware we flash is built from these files, so for the helm this beats the vendor's datasheet.	High
FW	Firmware source. Read from the ESC firmware (AM32) source at the commit below. Excerpt archived.	High
WEB	Vendor spec (web). Vendor or retailer specification seen through web search. Not yet checked against an archived PDF.	Medium
KN	Known datasheet value. A well-known datasheet figure quoted from memory. Must be checked against the archived PDF before it is relied on.	Medium
EST	Engineering estimate. Calculated or assumed. Replaced by a measurement at the stage shown.	Low
TEST	Measure. No trustworthy datasheet exists (salvaged, generic or water-loaded parts). The bench test is the datasheet.	Pending

Vendor sites (and ardupilot.org) were not reachable from the environment this list was written in; GitHub was. So ArduPilot and AM32 sources are archived in docs/kcl/sources/, and vendor datasheets are listed in section 9 with a script to fetch them.

2. Findings

CR-04 accepted (Issue B). The baseline flight controller (Matek F405-TE) needed 9 V, and our 3S Li-ion pack sags below that under load near empty. The boat could have lost its autopilot during the very return-home that the low-battery failsafe starts. The SpeedyBee F405 WING APP runs from 7 V, keeps microSD logging, costs about £17 less, and is now the baseline (ADD DD-19). Section 5 has the numbers.

ID	Finding	Type	Goes to
KF-01	The baseline flight controller browns out near empty. The Matek F405-TE needs at least 9 V in. Our 3S Li-ion pack is at 9.6 V (resting) when the critical battery failsafe fires, and it sags under load: with acceptance-limit cells the pack resistance is about 0.22 Ω . So the F405-TE would reset during the return-home it is flying, above about 1.4 A of motor current (section 5). PWR-D12 already requires supplies to hold at 9.0 V, which the F405-TE cannot meet. The SpeedyBee F405 WING APP runs from 7 V, has microSD logging, 12 DShot outputs and an ArduPilot board definition, and costs about £17 less.	Decision	CR-04 accepted (Issue B): KC-01 is the SpeedyBee F405 WING APP (ADD DD-19)
KF-02	Limit motor power so sag can't brown out anything. Even with the SpeedyBee, full thrust on a nearly flat, high-resistance pack could pull the rail below the ESCs' 7.2 V minimum. ArduPilot's BATT_WATT_MAX (Rover) reduces maximum throttle when power exceeds a limit. Set to 70 W, it keeps the rail above 7.5 V at the critical failsafe voltage while still leaving about 2 N of thrust, against 1.1 N needed for ENV-002.	Change	HLM parameter baseline (section 7)
KF-03	The arming-key MOSFET must switch the positive side. ICD IF-06 says 'gate pulled down → rail off', which reads like a low-side N-MOSFET, the kind of cheap trigger module sold for this. Switching the ESC negative while its signal ground is tied to the flight controller lets motor current find a way back through the signal wires, which can destroy the FC's output pins. So the key switch shall be a high-side P-MOSFET, with a gate RC to soft-start the ESC input capacitors (2 × 470 μ F).	Change	ICD IF-06, SSS-PWR PWR-D07
KF-04	The flight controller's built-in wireless is a second, uncontrolled command path. The SpeedyBee stack includes a Wi-Fi/Bluetooth board wired to UART6 and set up for MAVLink by default. That would let any phone near the boat talk to the autopilot, bypassing IF-01/IF-02 and the adult PIN. Don't fit the wireless board, and set SERIAL6_PROTOCOL = -1 as well.	Change	HLM parameter baseline, ICD IF-02
KF-05	Two microSD cards, not one. The Pi Zero 2W boots from one card; the flight controller logs to another (DD-15). The ADD BOM counted one. +£4.	Change	ADD BOM
KF-06	The boat link must be 2.4 GHz. The Pi Zero 2W's radio is 2.4 GHz only. The bank access point must run on 2.4 GHz (channel 1, 6 or 11). MT7610U/MT7612U USB adapters have in-kernel (mt76) drivers with access-point mode on Raspberry Pi OS, so no out-of-tree driver is needed.	Change	ICD IF-01
KF-07	The simulator needs our own boat physics. ArduPilot's built-in motorboat model gives 50 N at full throttle (hull drag equal to thrust at 10 m/s). Ours is roughly 4 N at 1.5 m/s. Simulated failsafe and weed tests would be meaningless at that scale. SITL has a JSON interface for an external physics model, so we write a small Python boat model parameterised from this list (section 8).	Change	Proposed DD-19; SSS-SIM, ICD IF-21
KF-08	Turn off the ESC's own low-voltage cut-off. AM32 has a configurable low-voltage cut-off that stops the motor after 10 s below its threshold. On a Li-ion pack it could strand the boat while ArduPilot is still trying to bring it home. Disable it in the ESC settings; battery protection belongs to the helm (FS-001) and the BMS.	Change	SSS-PRP ESC configuration
KF-09	Free motor-RPM telemetry (opportunity). AM32 supports bidirectional DShot, which reports each motor's RPM to ArduPilot. That gives B7 a direct 'motor dead / prop fouled' signal instead of inferring it from track divergence (FM-16, FM-27). It needs the motors on outputs 1 and 4, the only BIDIR-capable output on each of the first two timer groups. Not in the baseline; worth a bench trial.	Note	Candidate for HLM Issue D
KF-10	Calibrate the battery monitor on the bench. ArduPilot's board file and its README disagree on the voltage scale (11.05 against 11.5), which is a 4% difference and outside the $\pm 1\%$ of PWR-D13. The current scale (50 A/V) depends on the PDB fitted. Both get calibrated against a multimeter (IF-07 test).	Note	IF-07 bench test

ID	Finding	Type	Goes to
KF-11	The Pi 5 has no analogue audio output. The speaker must be USB (or a USB sound card), not a 3.5 mm jack speaker.	Change	MCN purchasing

3. The list

KC-08 and KC-13 are not used: the main switch and the moisture traces turned out not to feed any software value beyond what the specifications already say.

ID	Component	Qty	Proposed part	Status
KC-01	Flight controller (helm) (HLM)	1	SpeedyBee F405 WING APP (FC + PDB; wireless board not fitted)	Baseline (CR-04 accepted)
KC-02	GNSS receiver + compass (HLM)	1	M10 GNSS with QMC5883L compass on one board (e.g. Matek/UMT M10Q-5883 or Foxeer M10Q-250-5883)	Proposed
KC-03	Electronic speed controller (×2) (PRP)	2	AM32 single ESC, 20 A, 2-6S (e.g. OddityRC AM32 6S 20A Nano)	Proposed
KC-04	Thruster motor (×2) (PRP)	2	2205-2207 FPV outrunner, 1700-2300 KV, 5 mm shaft, run flooded	Class only
KC-05	Battery cells (×3) (PWR)	3	Salvaged 18650 Li-ion accepted to PWR-D20	Baseline (CR-03)
KC-06	Battery management system (PWR)	1	3S Li-ion BMS, 20-25 A, with balancing	Class only
KC-07	Arming-key switch (PWR)	1	Glass reed switch (normally open) driving a high-side P-MOSFET (AOD4185 class) on a small perfboard	Proposed
KC-09	5 V buck for the mission computer (PWR)	1	Synchronous buck module, 5.1 V, ≥ 3 A continuous	Class only
KC-10	Mission computer (MCP)	1	Raspberry Pi Zero 2 W	Baseline
KC-11	Camera (MCP)	1	OV5647 5 MP module with Pi Zero ribbon (Camera Module v1 compatible)	Baseline
KC-12	Box temperature sensor (MCP)	1	DS18B20 (TO-92)	Baseline (FMEA A-11)
KC-14	LED beacon (REC)	1	WS2812B addressable LEDs (4-8) on the FC LED pad	Baseline
KC-15	Bank Wi-Fi adapter (MCN)	1	USB adapter on MT7610U or MT7612U chipset with external antenna	Proposed
KC-16	Mission Control computer (MCN)	1	Raspberry Pi 5 (owned)	Baseline
KC-17	Microphone and speaker (MCN)	1	USB microphone (16 kHz mono capable) and USB speaker	Class only
KC-18	Panel buttons (×4) (MCN)	4	30 mm arcade buttons with 5 V LED	Class only
KC-19	microSD cards (×2) (MCP)	2	32 GB A1/U1 (one for the Pi Zero, one for the flight controller)	Proposed (KF-05)

Motor rail is live only with the key in; the FC and mission computer stay powered with it out.

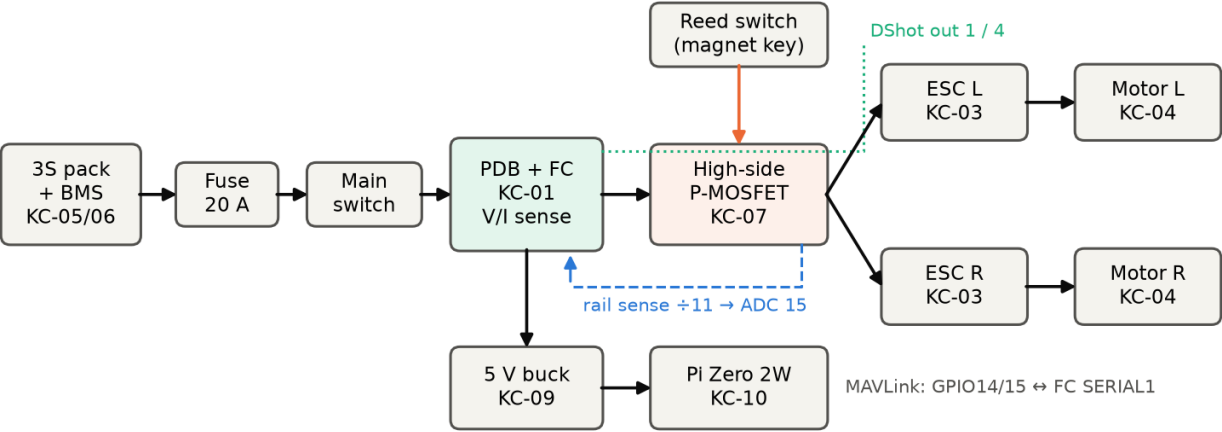


Figure 1. Boat power path with the proposed parts. The arming key switches the positive side of the motor rail (KF-03), and the helm senses that rail on its airspeed ADC pad.

4. Cost reconciliation

With the proposed parts the estimate is **£181**, against the £185 cap and £180 target (CON-001). Cheaper flight controller, realistic ESC price and a second microSD card roughly cancel out. The conditional antenna pole kit sits in the field-kit budget, outside the cap (CR-08).

Group	Item	£ (est.)	KC	Change from ADD Issue E
Boat	Flight controller: SpeedyBee F405 WING APP (FC + PDB)	45	KC-01	–£17 (was Matek F405-TE £62; CR-04)
Boat	M10 GNSS + QMC5883L compass	14	KC-02	-
Boat	Pi Zero 2W	15	KC-10	-
Boat	OV5647 camera + Zero cable	8	KC-11	-
Boat	microSD 32 GB × 2	8	KC-19	+£4 (KF-05)
Boat	2 × 2205-class motor	12	KC-04	-
Boat	2 × AM32 20 A ESC	22	KC-03	+£12 (was £10; realistic price)
Boat	3 × tested salvaged 18650 + 3S BMS	7	KC-05, KC-06	-
Boat	Box, PG7 glands, fuse	6	-	-
Boat	Foam, hi-vis, WS2812B beacon	5	KC-14	-
Boat	5 V 3 A buck	3	KC-09	-
Boat	Reed switch + high-side P-MOSFET switch	4	KC-07	Same cost, different circuit (KF-03)
Boat	IP67 main power switch	4	-	-
Boat	DS18B20 box temperature sensor	2	KC-12	-
Bank	4 arcade buttons (5 V LED)	6	KC-18	-
Bank	USB mic + USB speaker	8	KC-17	Speaker must be USB (KF-11)
Bank	USB Wi-Fi adapter, MT7610U/MT7612U	12	KC-15	-
Total (excluding conditional)		181		ADD Issue F baseline £181 (Issue E: £182)
Bank*	2 m pole + 3 m USB extension (only if V-08 needs it)	8	-	Unchanged

Prices are estimates from search results and earlier work; UK retail sites couldn't be opened from here. Confirm at order time. If the ESCs come in dear, BLHeli_S ESCs with Bluejay firmware save about £8.

5. Supply sag analysis

Pack resistance at the acceptance limits: $3 \times 60 \text{ m}\Omega$ cells + $15 \text{ m}\Omega$ BMS + $25 \text{ m}\Omega$ fuse, switch and wiring = **0.22 Ω** . The critical battery failsafe (BATT_CRT_VOLT) fires at 9.6 V resting, and the boat must then drive home on what's left.

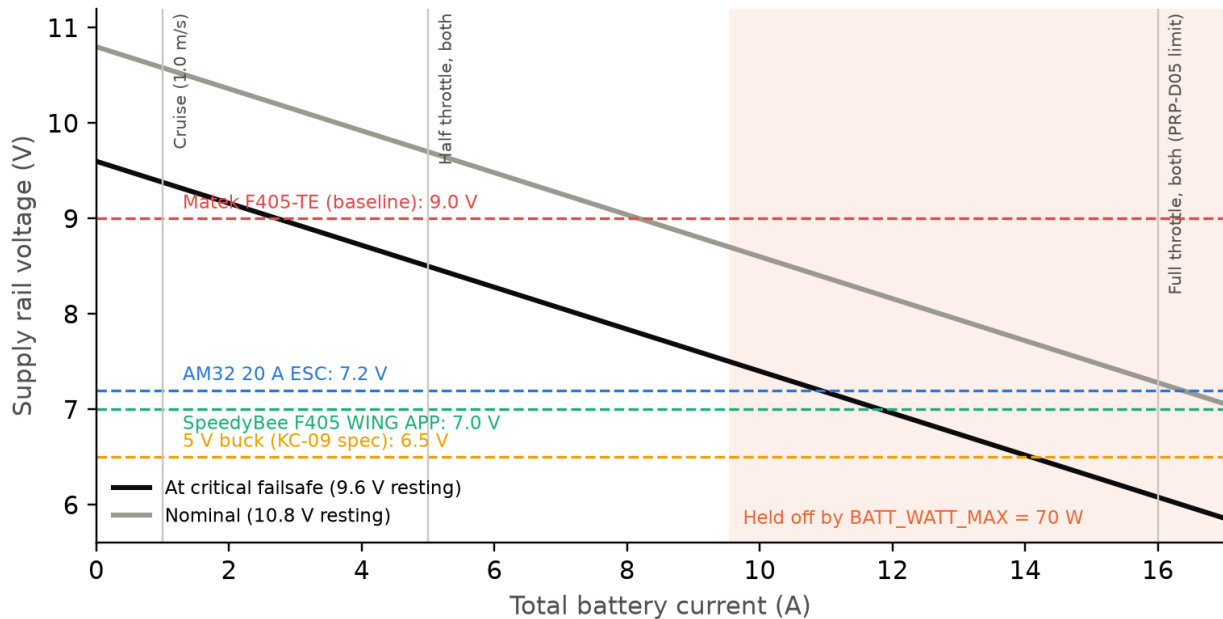


Figure 2. Rail voltage against battery current. The F405-TE (red) browns out above about 1.4 A near empty; cruise alone is about 1 A.

Supply	Minimum input	Source	Max current at 9.6 V resting (+0.3 V margin)
Matek F405-TE (baseline)	9.0 V	WEB	1.4 A
SpeedyBee F405 WING APP	7.0 V	WEB	10.5 A
AM32 20 A ESC	7.2 V	WEB	9.5 A
5 V buck (KC-09 spec)	6.5 V	EST	12.7 A

Power limit. BATT_WATT_MAX = 70 W keeps the rail above the highest of the remaining minimums (the ESC's 7.2 V) at the critical voltage. By momentum theory ($T \propto P^{2/3}$) that still leaves about 2.1 N, against 1.1 N needed to make headway in the ENV-002 wind. On a full pack the limit rarely acts. Whether it is enough gets tested on L2 with a programmable load (PWR-D12).

6. Component data

KC-01 Flight controller (helm)

Part	SpeedyBee F405 WING APP (FC + PDB; wireless board not fitted)
Alternatives	Baseline: Matek F405-TE (fails KF-01). Matek F405-WTE runs from 6.8 V but is end-of-life.
Why it's key	Every helm parameter, UART, ADC and output assignment comes from its board definition.
Traces	V-01, IF-04, IF-05, IF-07, IF-09, IF-22, DD-15, PWR-D12, PWR-D13

Parameter	Value	Src	Used by
MCU	STM32F405, 168 MHz, 1 MB flash	AP	V-01
Input voltage	2-6S; 7-36 V	WEB	PWR-D12, KF-01
5 V BEC	5.2 V, 2.4 A continuous / 3 A peak	WEB	IF-07, IF-22, IF-09
IMU / baro	ICM42688-P (SPI) / SPL06 (I2C 0x76)	AP	HLM
Compass	None on board; probes external I2C	AP	IF-22
Logging	microSD on SPI3 (FATFS); LOG_BACKEND_TYPE = 1	AP	LOG-001, DD-15
UART map	SERIAL1 USART1 (DMA) · SERIAL2 USART2 (SBUS inverter) · SERIAL3 USART3 (GPS) · SERIAL4 UART4 · SERIAL5 UART5 · SERIAL6 USART6 (wireless board)	AP	IF-04, IF-22
Companion UART	SERIAL1 (USART1, DMA both ways): Pi Zero MAVLink	AP	IF-04
PWM outputs	12, all DShot. Groups 1-2 (TIM4), 3-4 (TIM3), 5-7 (TIM8), 8-10 (TIM2), 11-12 (TIM1). BIDIR on outputs 1 and 4. Output 12 = LED pad	AP	IF-05, IF-09
Battery monitor	BATT_VOLT_PIN 10, BATT_CURR_PIN 11; scale 11.05 (hwdef) or 11.5 (README), 50 A/V: calibrate	AP	IF-07, KF-10
Spare ADC	RSSI pad = ADC pin 14, AIRSPD pad = pin 15; 3.3 V full scale, no on-board divider	AP	IF-06
Current sensor	90 A continuous, 215 A peak	AP	IF-07
Mass	≈ 25 g FC + PDB (12 g allocated)	EST	MEC-002

KC-02 GNSS receiver + compass

Part	M10 GNSS with QMC5883L compass on one board (e.g. Matek/UMT M10Q-5883 or Foxeer M10Q-250-5883)
Alternatives	Any u-blox M10 + ArduPilot-supported compass (QMC5883L/P, IST8310)
Why it's key	Sets position noise, update rate and the compass driver, address and orientation parameters.
Traces	IF-22, NAV-008, V-16

Parameter	Value	Src	Used by
Receiver	u-blox M10 (SAM-M10Q), GPS + Galileo + BeiDou + GLONASS	WEB	NAV-008
Update rate	5 Hz UBX with four constellations; ArduPilot configures it (GPS_TYPE auto)	WEB	HLM
Default baud	9600; ArduPilot auto-bauds and reconfigures	WEB	IF-22
Horizontal accuracy	≈ 1.5 m CEP open sky	KN	SIM model
Supply	4-9 V; ≈ 13 mA (receiver)	WEB	IF-22
Compass	QMC5883L on I2C, address 0x0D; ArduPilot probes it as external	AP	IF-22, TBC-09
Temperature range	−20 to 80 °C	WEB	-

KC-03 Electronic speed controller (×2)

Part	AM32 single ESC, 20 A, 2-6S (e.g. OddityRC AM32 6S 20A Nano)
Alternatives	BLHeli_S ESC flashed with Bluejay ($\approx$ £7 each; SERVO_DSHOT_ESC = 2). AM32 preferred: maintained, configurable, RPM telemetry.
Why it's key	Its protocol, reversing, signal-loss and low-voltage behaviour are failsafe behaviour. They are modelled in SITL and tested on L2.
Traces	IF-05, PRP-D05, PRP-D06, PRP-D07, PRP-D08, V-09

Parameter	Value	Src	Used by
Protocol	DSHOT300/600 and PWM	WEB	IF-05
Reversible	3D mode via SERVO_BLM_3DMASK (ArduPilot configures AM32 at boot)	AP	IF-05, PRP-D01
Voltage	7.2-25.2 V (2-6S)	WEB	KF-02
Current	20 A continuous, 30 A burst (10 s)	WEB	PRP-D05
Signal loss, armed	All phases off and reset after 0.5 s	FW	PRP-D07, FS-008, V-09
Re-arm	Needs > 1 s of zero command before it will drive again	FW	PRP-D08
Low-voltage cut-off	Configurable; stops after 10 s below threshold → disable (KF-08)	FW	FS-001
Telemetry	Bidirectional DSHOT RPM (optional, KF-09)	WEB	FM-16
BEC	5 V 1 A: leave unconnected (don't parallel with the FC BEC)	WEB	IF-07
Mass	2.6 g board; 36 × 24 mm	WEB	MEC-002

KC-04 Thruster motor (×2)

Part	2205-2207 FPV outrunner, 1700-2300 KV, 5 mm shaft, run flooded
Alternatives	Sealed ROV thruster motors (much dearer; future option)
Why it's key	Thrust and current against throttle are the most important simulator inputs, and no datasheet covers a small motor turning a printed prop in water.
Traces	PRP-D02, PRP-D05, IF-06

Parameter	Value	Src	Used by
Stator / KV	2205-2207 / 1700-2300 KV (class)	EST	PRP-D02
Static thrust, each	≥ 2.0 N forward, ≥ 0.75 N astern (PRP-D02 split)	EST	SIM model
Thrust vs throttle	Assume $T \propto \text{throttle}^2$ until measured	EST	SIM model
Current at full throttle	≤ 8 A each (PRP-D05 limit)	TEST	IF-06, TBC-06
Spin-up time constant	≈ 0.1 s	EST	SIM model

KC-05 Battery cells (×3)

Part	Salvaged 18650 Li-ion accepted to PWR-D20
Alternatives	New cells (e.g. 3000 mAh, 15 A class) if too few pass
Why it's key	Capacity and internal resistance set the failsafe thresholds, endurance and supply sag.
Traces	PWR-D01, PWR-D20, PWR-D21

Parameter	Value	Src	Used by
Capacity	≥ 2500 mAh each at 1 A (acceptance)	TEST	PWR-D20, BATT_CAPACITY
Internal resistance	≤ 60 m Ω each (acceptance); matched within 10 m Ω	TEST	KF-01, KF-02
Voltage	4.2 V full, 3.6 V nominal, 2.5-3.0 V empty per cell	KN	FS-001

Parameter	Value	Src	Used by
Pack	3S1P: 12.6 V full, 10.8 V nominal	EST	IF-06

KC-06 Battery management system

Part	3S Li-ion BMS, 20-25 A, with balancing
Alternatives	-
Why it's key	Its over-discharge cut-off is the last line after the helm failsafes; it must never act before them.
Traces	PWR-D02

Parameter	Value	Src	Used by
Over-discharge	2.5-2.8 V/cell (PWR-D02)	TEST	FS-001
Over-current trip	≤ 25 A	TEST	PWR-D02
Series resistance	≈ 15 m Ω (in the sag budget)	EST	KF-01

KC-07 Arming-key switch

Part	Glass reed switch (normally open) driving a high-side P-MOSFET (AOD4185 class) on a small perfboard
Alternatives	Low-side trigger modules are not acceptable (KF-03)
Why it's key	Its state is what the helm senses (BATT2) to allow arming. Its switching time and inrush are part of L2 tests.
Traces	IF-06, PWR-D07, PWR-D08, PWR-D09, V-13

Parameter	Value	Src	Used by
MOSFET	P-channel, $V_{DS} \geq 30$ V, $I_D \geq 30$ A, $R_{DS(on)} \leq 15$ m Ω at $V_{GS} = -10$ V	EST	IF-06
Gate drive	Reed closes $\rightarrow$ gate pulled toward ground via divider, V_{GS} clamped -10 V; open $\rightarrow 10$ k Ω pulls gate to source (off, fails safe)	EST	PWR-D07
Soft start	Gate RC ≈ 10 ms to limit inrush into 2×470 μ F	EST	KF-03
Rail sense	Divider 10 k Ω / 1 k Ω from the switched rail to the AIRSPD pad (ADC pin 15)	EST	PWR-D08, V-13

KC-09 5 V buck for the mission computer

Part	Synchronous buck module, 5.1 V, ≥ 3 A continuous
Alternatives	Pololu D36V28F5 (dearer, well documented)
Why it's key	Its minimum input voltage decides whether the mission computer survives sag. It is part of the sag analysis.
Traces	IF-08, PWR-D11, PWR-D12

Parameter	Value	Src	Used by
Output	5.1 V $\pm 2\%$, ≥ 3 A continuous	EST	IF-08
Minimum input	≤ 6.5 V at 1 A load	EST	KF-02
Ripple	≤ 50 mV p-p	EST	IF-08

KC-10 Mission computer

Part	Raspberry Pi Zero 2 W
Alternatives	-
Why it's key	Radio band, GPIO, 1-Wire and UART assignments feed IF-01, IF-03, IF-04.
Traces	MCP-D01, MCP-D03, MCP-D05, IF-01, IF-04

Parameter	Value	Src	Used by
SoC	RP3A0: 4 × Cortex-A53 at 1 GHz, 512 MB	KN	MCP-D01
Wi-Fi	2.4 GHz 802.11 b/g/n only	KN	IF-01, KF-06
UART to helm	GPIO14/15 (PL011 with Bluetooth disabled), 3.3 V	KN	IF-04
1-Wire	GPIO4 (w1-gpio overlay) for KC-12	KN	MCP-D04
Moisture input	GPIO with pull-up; traces short to ground when wet	EST	MCP-D03
Power	≈ 0.6 W idle, ≈ 3 W peak with camera	KN	MCP-D05

KC-11 Camera

Part	OV5647 5 MP module with Pi Zero ribbon (Camera Module v1 compatible)
Alternatives	Camera Module 3 (dearer, autofocus)
Why it's key	Field of view, resolution and capture latency set B2's photo geometry and timing.
Traces	MCP-D02, MCP-D10, MCP-D11

Parameter	Value	Src	Used by
Sensor	OV5647, 2592 × 1944; libcamera driver ov5647	KN	MCP-D10
Horizontal FOV	≈ 54° (standard lens)	KN	MCP-D02
Still capture latency	≈ 0.3-1 s in software	TEST	MCP-D10

KC-12 Box temperature sensor

Part	DS18B20 (TO-92)
Alternatives	-
Why it's key	B6 raises RTL at 60 °C box temperature from this reading.
Traces	A-11, MCP-D15

Parameter	Value	Src	Used by
Accuracy	± 0.5 °C from −10 to +85 °C	KN	A-11
Conversion time	750 ms at 12-bit	KN	MCP-D15
Interface	1-Wire, 3.0-5.5 V, 4.7 kΩ pull-up	KN	MCP-D04

KC-14 LED beacon

Part	WS2812B addressable LEDs (4-8) on the FC LED pad
Alternatives	-
Why it's key	The helm drives it, so the notify-LED parameters and the output assignment depend on it.
Traces	IF-09, REC-004

Parameter	Value	Src	Used by
Drive	Output 12 (LED pad), SERVO12_FUNCTION = 120 (NeoPixel1)	AP	IF-09
Supply / data	5 V; 3.3 V data works on most WS2812B, marginal on some (TBC-08)	KN	IF-09

Parameter	Value	Src	Used by
Current	≈ 60 mA per LED at full white	KN	IF-07

KC-15 Bank Wi-Fi adapter

Part	USB adapter on MT7610U or MT7612U chipset with external antenna
Alternatives	Pi 5 on-board Wi-Fi (shorter range; no external antenna)
Why it's key	It must run as an access point on 2.4 GHz with an in-kernel driver.
Traces	IF-01, MCP-D16

Parameter	Value	Src	Used by
Driver	mt76x0u / mt76x2u, in-kernel since Linux 4.19	WEB	IF-01
AP mode	Supported (hostapd)	WEB	IF-01
Band used	2.4 GHz, channel 1/6/11 (Pi Zero 2W limit)	KN	KF-06

KC-16 Mission Control computer

Part	Raspberry Pi 5 (owned)
Alternatives	-
Why it's key	Hosts C1-C10. GPIO for the panel, USB for audio and Wi-Fi.
Traces	IF-12

Parameter	Value	Src	Used by
Audio	No analogue output: USB speaker required (KF-11)	KN	IF-12
Power	≈ 5 W average, ≈ 8 W peak (STT)	EST	MC-006

KC-17 Microphone and speaker

Part	USB microphone (16 kHz mono capable) and USB speaker
Alternatives	USB sound card + small powered speaker
Why it's key	On-device STT/TTS sample rate and device names.
Traces	NLI-002

Parameter	Value	Src	Used by
Mic format	16 kHz, 16-bit mono (resampled if needed)	EST	NLI-002
Speaker	USB audio class, ≥ 85 dB at 1 m	EST	NLI-003

KC-18 Panel buttons (×4)

Part	30 mm arcade buttons with 5 V LED
Alternatives	12 V-LED versions need a different driver: avoid
Why it's key	Debounce timing and LED drive feed IF-12.
Traces	IF-12, MC-002

Parameter	Value	Src	Used by
Switch	Normally-open microswitch; bounce ≤ 10 ms	EST	IF-12
LED	5 V, ≈ 20 mA, driven through N-MOSFET	EST	IF-12

KC-19 microSD cards (×2)

Part	32 GB A1/U1 (one for the Pi Zero, one for the flight controller)
Alternatives	-
Why it's key	The FC needs FAT32 (≤ 32 GB). The Pi reserves ≥ 5 GB for photos.
Traces	DD-15, MCP-D13

Parameter	Value	Src	Used by
FC card	≤ 32 GB, FAT32	AP	DD-15
Pi card	32 GB, A1 or better; read-only root	EST	MCP-D13

7. Helm allocation and parameter changes

From the SpeedyBeeF405WING board definition (DS-01). These close TBC-02 and most of TBC-05 once CR-04 is accepted.

Function	FC resource	Notes
Mission computer (IF-04)	SERIAL1 = USART1 (DMA)	Pads R1/T1; 115200 8N1
GNSS (IF-22)	SERIAL3 = USART3	Board default GPS port
Compass (IF-22)	I2C1 (PB8/PB9)	QMC5883L at 0x0D
Left motor (IF-05)	Output 1 (TIM4, BIDIR)	DShot300, 3D
Right motor (IF-05)	Output 4 (TIM3, BIDIR)	DShot300, 3D. Output 2/3 unused
Beacon (IF-09)	Output 12 (LED pad, TIM1)	WS2812B
Battery V/I (IF-07)	ADC 10 / 11 via PDB	Calibrate
Motor-rail sense (IF-06)	ADC 15 (AIRSPD pad)	10 kΩ / 1 kΩ divider
Unused	SERIAL2, 4, 5; SERIAL6 disabled	Wireless board not fitted

7.1 Parameter baseline changes for SSS-HLM Issue D

Parameter	Proposed value	Was	Src	Refs
SERIAL1_PROTOCOL / SERIAL1_BAUD	2 (MAVLink 2) / 115	SERIAL2 (TBC-02)	AP	KC-01, IF-04
SERIAL2_PROTOCOL	-1 (unused; SBUS inverter on RX)	Companion	AP	KC-01
SERIAL3_PROTOCOL	5 (GPS)	-	AP	KC-02, IF-22
SERIAL6_PROTOCOL	-1 (wireless board not fitted)	Board default 2	AP	KF-04
SERVO1_FUNCTION / SERVO4_FUNCTION	73 ThrottleLeft / 74 ThrottleRight	SERVO1 / SERVO2	AP	KC-01, KF-09
MOT_PWM_TYPE	6 (DShot300)	DShot300 (TBC-05)	AP	KC-03, IF-05
SERVO_DSHOT_ESC	1 (AM32)	-	AP	KC-03
SERVO_BLH_3DMASK	9 (outputs 1 and 4 reversible)	-	AP	KC-03, IF-05
SERVO_BLH_RVMASK	Set on the bench so forward is forward	-	TEST	KC-03
MOT_REV_DELAY	0.2 s (tune on L2)	-	EST	KC-03, KC-04
BATT_VOLT_MULT / BATT_AMP_PERVLT	Calibrated (≈ 11.05 - 11.5 / 50)	Measured	AP	KF-10, IF-07
BATT_FS_VOLTSRC	1 (sag-compensated)	-	AP	KF-01
BATT_WATT_MAX / MOT_BAT_WATT_TC	70 W / 2 s	-	EST	KF-02
BATT2_MONITOR / BATT2_VOLT_PIN / BATT2_VOLT_MULT	3 (analogue voltage) / 15 (AIRSPD pad) / 11.0	Analogue V	AP	KC-07, IF-06
LOG_BACKEND_TYPE	1 (file on microSD)	File	AP	KC-01
NTF_LED_TYPES / SERVO12_FUNCTION	NeoPixel / 120 (LED pad)	SERVO3 NeoPixel1	AP	KC-14, IF-09
GPS_TYPE	1 (auto; u-blox configured by ArduPilot)	Auto	AP	KC-02
COMPASS_ORIENT / COMPASS_EXTERNAL	Per mounting / 1	Per module	AP	KC-02, IF-22

8. Simulator model parameters

ArduPilot's built-in simulated motorboat gives 50 N at full throttle, for a much bigger boat (KF-07). The simulator will run the real ArduPilot firmware against a small Python boat model through SITL's JSON interface (DS-18). These are its starting parameters. Most are estimates, and the last column says which measurement replaces each one; the model is only as good as those measurements.

Symbol	Meaning	Value	Units	Source	Replaced by
m	Mass (all-up)	1.48	kg	EST (ADD mass budget)	Weigh the build
m_a	Surge added mass	10% of m	kg	EST	Pool coast-down
I_zz	Yaw inertia, uniform box	0.063	kg·m ²	EST	Pool turn tests
y_pod	Pod lateral offset (HUL-D04 / 2)	±0.135	m	EST	CAD
T_fwd	Max forward thrust, each pod	2.0	N	EST (PRP-D02)	L3 bollard pull
T_ast	Max astern thrust, each pod	0.75	N	EST (PRP-D02)	L3 bollard pull
T(u)	Thrust law	$T_{\max} \cdot u \cdot u $	-	EST	Thrust stand
τ_m	Motor spin-up time constant	0.10	s	EST	Thrust stand
k_u	Surge drag coefficient ($F = k_u \cdot v \cdot v $)	1.62	N·s ² /m ²	EST (SSS-PRP §6)	Pool coast-down
k_r	Yaw damping ($M = k_r \cdot r \cdot r $)	0.05	N·m·s ²	EST	Pool turn tests
A_w, C_d	Windage area and drag coefficient	0.035, 1.1	m ² , -	EST (SSS-PRP §6)	-
P(T)	Electrical power per pod	$k_p \cdot T^{1.5}$, k_p set so cruise draws 8 W	W	EST	Thrust stand current
P_hotel	Hotel load	3.1	W	EST (ADD power budget)	Bench
C, R_pack	Pack capacity and resistance	2.5, 0.22	Ah, Ω	TEST (PWR-D20 limits)	Cell acceptance data
OCV(SoC)	Open-circuit voltage curve	Generic NMC 18650 table	V	KN	Discharge test
σ_{gnss} , f_{gnss}	GNSS noise and rate	1.0, 5	m, Hz	KN / WEB	Static log on the bank
V_min	Supply brown-out thresholds	FC 7.0, ESC 7.2, buck 6.5	V	WEB / EST	L2 sag test

A 3-DOF model is enough: surge, sway and yaw, with thrust from two pods, quadratic drag, windage, and the battery feeding back into available thrust. Waves, heel and pitch are left out: the catamaran is stiff (SSS-HUL section 6), and none of the SC tests depend on them.

9. Datasheet register

5 archived, 7 ready to fetch, the rest to locate or to measure. python3 docs/kcl/src/fetch_datasheets.py downloads the 'To fetch' entries into docs/kcl/datasheets/ and records their SHA-256, so a datasheet can't silently change under us. Run it anywhere with ordinary internet access.

ID	KC	Document	Status
DS-01	KC-01	ArduPilot board definition, SpeedyBeeF405WING (hwdef.dat, README, defaults, pinout, wiring) https://github.com/ArduPilot/ardupilot/tree/a11f7351a9bcd860cbdb55bbeb17a5196536ba23/libraries/AP_HAL_ChibiOS/hwdef/SpeedyBeeF405WING	Archived
DS-02	KC-01	SpeedyBee F405 WING APP user manual	To locate
DS-03	KC-01	ArduPilot board definition, MatekF405-TE, and the F405-TE family wiki page (for the comparison) https://github.com/ArduPilot/ardupilot/tree/a11f7351a9bcd860cbdb55bbeb17a5196536ba23/libraries/AP_HAL_ChibiOS/hwdef/MatekF405-TE	Archived
DS-04	KC-02	Matek M10Q-5883 product page and manual https://www.mateksys.com/?portfolio=m10q-5883	To fetch
DS-05	KC-02	u-blox SAM-M10Q data sheet	To locate
DS-06	KC-02	QST QMC5883L data sheet	To locate
DS-07	KC-03	OddityRC AM32 6S 20A Nano ESC product page https://oddityrc.com/products/am32-6s-20a-nano-esc	To fetch
DS-08	KC-03	AM32 firmware source: signal-loss and re-arm logic (excerpt) https://github.com/am32-firmware/AM32/blob/2738df3240baa5bd4295b460cf0c5cfe0bd49d97/Src/main.c	Archived
DS-09	KC-04	Motor + printed prop: thrust, current and RPM against throttle, in water	Measure
DS-10	KC-05	Cell acceptance records (PWR-D21)	Measure
DS-11	KC-06	BMS: trip thresholds measured on the bench	Measure
DS-12	KC-07	AOD4185 P-MOSFET data sheet	To locate
DS-13	KC-10	Raspberry Pi Zero 2 W product brief https://datasheets.raspberrypi.com/rpizero2/raspberry-pi-zero-2-w-product-brief.pdf	To fetch
DS-14	KC-11	Raspberry Pi camera documentation (OV5647 / Camera Module v1) https://www.raspberrypi.com/documentation/accessories/camera.html	To fetch
DS-15	KC-12	Analog Devices DS18B20 data sheet https://www.analog.com/media/en/technical-documentation/data-sheets/DS18B20.pdf	To fetch
DS-16	KC-15	MT7612U Linux support notes (morrownr/7612u) https://github.com/morrownr/7612u	To fetch
DS-17	KC-16	Raspberry Pi 5 product brief https://datasheets.raspberrypi.com/rpi5/raspberry-pi-5-product-brief.pdf	To fetch
DS-18	-	ArduPilot SITL JSON physics interface (readme) https://github.com/ArduPilot/ardupilot/tree/a11f7351a9bcd860cbdb55bbeb17a5196536ba23/libraries/SITL/examples/JSON	Archived
DS-19	-	ArduPilot SITL motorboat and skid-steer default parameters https://github.com/ArduPilot/ardupilot/tree/a11f7351a9bcd860cbdb55bbeb17a5196536ba23/Tools/autotest/default_params	Archived

'Measure' entries are deliberate: for salvaged cells, a generic BMS and a drone motor in water, no datasheet would be trustworthy. The bench result becomes the datasheet.

10. Where the changes went

Document	Where it went (Issue B)	Findings
ADD Issue F	Done: KC-01 in HLM, BOM (£181) and mass budget (DD-19); own boat physics via SITL JSON (DD-20); DD-15 unchanged (microSD kept).	KF-01, KF-05, KF-07
ICD Issue E	Done: IF-04 on SERIAL1 (TBC-02 closed); IF-05 outputs 1 and 4, AM32 3D DShot (TBC-05 closed, bench check with V-09); IF-06 high-side switch and AIRSPD-pad sense (TBC-07 closed); SERIAL6 off; IF-21 rewritten for the JSON model (TBC-03/04 closed). IF-01 was already 2.4 GHz.	KF-03, KF-04, KF-06, KF-07
SSS-HLM Issue D	Done: parameter baseline now generated from software/params; BATT_WATT_MAX (HLM-D43) and one command path (HLM-D44).	KF-02, KF-04
SSS-PWR Issue C	Done: PWR-D07 high-side; PWR-D22 supplies work down to 7.5 V under the power limit.	KF-01, KF-03
SSS-PRP Issue B	Done: AM32 ESC and its configuration (3D, LVC off, BEC unused; PRP-D16).	KF-08
SSS-SIM Issue D	Done: boat model from this list (SIM-D25); injections on the model side (SIM-D07).	KF-07
FMEA Issue D	Done: FM-08 (supply brown-out) re-rated with its real cause and closed by A-21. Still open: RPM telemetry as a detection control for FM-16/FM-50 (dead motor vs weed).	KF-01, KF-09

10.1 Measurements that matter most to the simulator

- **Thrust, current and RPM against throttle** for one pod in water (DS-09). A luggage scale and a bucket will do. This single curve sets the model's thrust, power and endurance.
- **Cell capacity and resistance** from acceptance testing (DS-10). These set R_pack and the sag margin.
- **ESC signal-loss stop time** on the bench (V-09). The AM32 source says 0.5 s; confirm on the actual ESC.
- **Coast-down and turn tests** in the pool to replace the drag and inertia estimates.

Appendix A. Archived sources

Source	Repository and commit
ArduPilot firmware (board definitions, SITL)	ArduPilot/ardupilot @ a11f7351a9bc
ArduPilot wiki (F405-TE family page)	ArduPilot/ardupilot_wiki @ 5365bb696d91
AM32 ESC firmware (signal-loss excerpt)	am32-firmware/AM32 @ 2738df3240ba

ArduPilot and AM32 are GPLv3; the archived files are unmodified copies or excerpts, kept for reference.